

Задание 1:

--1 Задание

```
delete from EmployeeProjects
where EmployeeID = 4;
```

```
delete from Employees
where EmployeeID = 4;
```

-- Вставка новых сотрудников

```
insert into Employees (FirstName, LastName, Department, Salary) values
('Nikita', 'Millerk', 'Finance', 67000.00),
('Nikola', 'Lemp', 'IT', 72000.00);
```

-- Выборка всех сотрудников

```
select * from Employees;
```

-- Выборка сотрудников из отдела IT

```
select FirstName, LastName from Employees
where Department = 'IT';
```

-- Обновление зарплаты Alice Smith

```
update Employees
set Salary = 65000.00
where FirstName = 'Alice' and LastName = 'Smith';
```

-- Удаление сотрудника с фамилией Prince

```
delete from Employees
where LastName = 'Prince';
```

-- Финальная проверка

```
select * from Employees;
```

[DROP TABLE IF EXISTS EmployeeProjects; DF | Фильтрация данных не поддерживается

	123 employeeid	A-Z firstname	A-Z lastname	A-Z department	123 salary
1	1	Alice	Smith	HR	60 000
2	2	Bob	Johnson	IT	75 000
3	3	Charlie	Brown	Finance	62 000
4	5	Eve	Davis	HR	58 000
5	6	Nikita	Millerk	Finance	67 000
6	7	Nikola	Lemp	IT	72 000

Рисунок 1 – добавлен Nikita and Nikola


```
update Employees set Email = 'eve.davis@example.com' where FirstName = 'Eve' and
LastName = 'Davis';
update Employees set Email = 'nikita.millerk@example.com' where firstname =
'Nikita' and lastname = 'Millerk';
update Employees set Email = 'nikola.lemp@example.com' where FirstName = 'Nikola'
and lastname = 'Lemp';

alter table Employees
add constraint unique_email unique (Email);

alter table Departmens
rename column location to OfficeLocation;
```

[illegible]

Задание 3:

[illegible]

Задание 4:

```
select 3, projectID, 120 from new_proj;
commit; -- фиксируем изменения
```

```
select employeeID, FirstName, LastName, Department, Salary
from employees
order by EmployeeID ASC;
```

employees 1 employees 1 (2) employees 1 (3) departmens 1 (4) employees 1 (5) employ					
DROP TABLE IF EXISTS EmployeeProjects; DF Фильтрация данных не поддерживается					
	123 employeeid	A-Z firstname	A-Z lastname	A-Z department	123 salary
1	1	Alice	Smith	Senior IT	71 500
2	2	Bob	Johnson	Senior IT	75 000
3	3	Charlie	Brown	Finance	62 000
4	5	Eve	Davis	HR	63 800
5	7	Nikola	Lemp	Senior IT	72 000

Задание 5:

```
-- Создание функции для расчёта годового бонуса (10% от зарплаты)
create or replace function CalculateAnnualBonus(emp_id INT, emp_salary DECIMAL)
returns DECIMAL as $$
BEGIN
    return emp_salary * 0.10;
END;
$$ language plpgsql;
```

```
-- Вывод потенциального бонуса для каждого сотрудника
select
    EmployeeID,
    FirstName,
    LastName,
    Salary,
    CalculateAnnualBonus(EmployeeID, Salary) AS Bonus
from Employees;
```

```
-- Создание представления для сотрудников из отдела 'IT'
create or replace VIEW IT_Department_View as
SELECT
    EmployeeID,
    FirstName,
    LastName,
    Salary,
    Department
from Employees
where Department = 'IT';
```

```
-- Получение данных из представления IT_Department_View
select * from IT_Department_View;
```



```

update Projects
set EndDate = StartDate + interval '1 year'
where EndDate is null;

begin;

-- Вставка нового сотрудника с возвратом его ID
with new_employee as(
    insert into Employees (FirstName, LastName, Department, Salary)
    values ('Pavel', 'Golub', 'IT', 50000.00)
    returning EmployeeID
),
target_project as(
    select ProjectID from Projects where ProjectName = 'Website Redesign'
)
-- Назначение сотрудника на проект
insert into EmployeeProjects (EmployeeID, ProjectID, HoursWorked)
select ne.EmployeeID, tp.ProjectID, 80
from new_employee ne, target_project tp;

commit;

--Проверка: проекты Bob Johnson с >150 часов
select * from Projects
where ProjectName in(
    select Projects.ProjectName
    from Projects
    join EmployeeProjects on Projects.ProjectID = Employeeprojects.ProjectID
    join Employees on employeeprojects.EmployeeID = Employees.EmployeeID
    where Employees.FirstName = 'Bob' and Employees.LastName = 'Johnson'
    and Employeeprojects.HoursWorked > 150
);

--Проверка: бюджеты проектов
select ProjectName, Budget from Projects;

-- Проверка: даты завершения проектов
select ProjectName, StartDate, EndDate from Projects;

-- Проверка нового сотрудника и его назначение
select Employees.EmployeeID, Employees.firstname e, Employees.LastName,
Employeeprojects.ProjectID, Employeeprojects.HoursWorked
from Employees
join EmployeeProjects on Employees.EmployeeID = employeeprojects.EmployeeID
where Employees.FirstName = 'Pavel' and Employees.LastName = 'Golub';

```

